

REQUIREMENT ANALYSIS

AI-Assisted Smart Grievance Redressal System

1. Project Overview

A centralized organizational grievance management system that allows users to submit and track grievances while helping officers process them using AI-assisted analysis.

The system focuses on:

- Grievance management
- AI-based classification
- Priority recommendation
- Similar grievance detection

The AI provides recommendations, while the officer makes the final decision.

2. Problem Statement

Traditional grievance systems mainly handle submission, tracking, assignment, and resolution. As grievance volume increases, manual categorization, prioritization, and identification of duplicate complaints become time-consuming.

The proposed system uses AI to reduce this manual effort and assist officers in handling grievances more efficiently.

3. Objectives

- Centralize grievance submission and tracking.
- Automatically recommend grievance categories.
- Recommend grievance priority.
- Identify similar/duplicate grievances.
- Maintain grievance history.
- Provide role-based access.
- Provide dashboards and reports.
- Keep human officers responsible for final decisions.

4. Scope

The system is designed for a general organization with departments such as IT, HR, Finance, Facilities, and Administration.

It includes:

- User authentication
- Grievance submission and tracking
- AI-assisted grievance analysis

- Officer assignment and resolution
- Escalation
- Notifications
- Administration
- Reports and dashboards

5. Stakeholders & Roles

Role	Main Responsibilities
User	Submit and track grievances
Grievance Officer	Review, assign, update and resolve grievances
Administrator	Manage users, departments and categories
Management	Monitor statistics and performance

6. Grievance Workflow

Submitted

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Under Review

↓

Assigned

↓

In Progress

↓

Resolved

↓

Closed

If required:

In Progress → Escalated → In Progress

7. Functional Requirements

FR-01 — Authentication

Users shall log in and access features according to their roles.

FR-02 — Grievance Submission

Users shall submit grievances and receive a unique grievance ID.

FR-03 — Grievance Tracking

Users shall view their grievance status and updates.

FR-04 — AI Classification

The system shall recommend a grievance category based on its description.

FR-05 — Priority Recommendation

The system shall recommend Low/Medium/High/Critical priority.

FR-06 — Similar Grievance Detection

The system shall identify potentially related or duplicate grievances.

FR-07 — Assignment & Status

Officers shall assign, update, escalate and resolve grievances.

FR-08 — History

The system shall maintain grievance activity and status history.

FR-09 — Administration

Administrators shall manage users, departments and categories.

FR-10 — Reports

Management shall view grievance statistics and trends.

8. AI Requirements

The AI module will contain three core components:

1. Classification

Grievance Text → Category

Example:

"Office Wi-Fi is not working." → IT / Network

2. Priority Prediction

Grievance + Relevant Factors → Priority

→ Low / Medium / High / Critical

3. Similarity Detection

New Grievance → Compare with Existing Grievances → Similar Cases

AI recommendations can be accepted or overridden by the officer.

9. Dataset

Since a suitable organizational grievance dataset may not be available, we will create a curated/synthetic dataset containing:

- Grievance description
- Category

- Priority
- Department
- Other relevant attributes

The dataset will be divided into training and testing data for AI evaluation.

10. Non-Functional Requirements

- **Security:** Role-based access and protected grievance data.
- **Performance:** Fast response under expected usage.
- **Reliability:** Accurate and consistent data storage.
- **Scalability:** Support increasing users and grievances.
- **Usability:** Simple interface for non-technical users.
- **Maintainability:** Modular system architecture.

11. Business Rules

- Every grievance receives a unique ID.
- Users can access only their own grievances.
- Only authorized officers can modify grievance status and resolution.
- AI recommendations do not automatically become final decisions.
- Officers can override AI recommendations.
- Important grievance changes are recorded in history.
- Grievances follow the defined lifecycle.

12. Technology Stack

Frontend

- ASP.NET Core MVC
- Razor
- HTML/CSS/JavaScript
- Bootstrap
- Chart.js

Backend

- C#
- ASP.NET Core Web API
- Entity Framework Core

Database

- SQL Server
- SSMS

AI/ML

- Python
- Pandas
- NumPy
- Scikit-learn
- NLP/embedding techniques

Tools

- Visual Studio 2022
- Git & GitHub

13. Main Modules

1. Authentication & Authorization
2. Grievance Management
3. AI-Assisted Triage
4. Administration
5. Dashboard & Reporting

14. What Makes It Different?

Existing systems primarily focus on:
Submit → Track → Assign → Resolve

Our system adds:

AI-assisted classification + priority recommendation + similar grievance detection

So the project is not simply another grievance CRUD application. It is a decision-support system that combines grievance management with AI-assisted triage, while keeping the final decision with the human officer.

15. High-Level Architecture

User

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ASP.NET Core MVC

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ASP.NET Core Web API

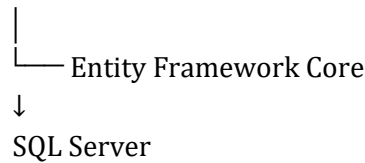
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Business Logic

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├── AI/ML Module
│   ├── Classification
│   ├── Priority
│   └── Similarity

```



16. Database Schema (Implementation Details)

To support the above requirements, the SQL Server database will require the following foundational tables, including primary entity tables and junction tables to represent many-to-many relationships.

Primary Tables

1. Users (User Management)

- **Description:** Stores all system users including employees, grievance officers, and administrators.
- **Key Columns:** UserID (PK), FullName, Email, PasswordHash, RoleID (FK), DepartmentID (FK), CreatedAt, IsActive.

2. Roles (Role-Based Access Control)

- **Description:** Defines the roles available in the system (e.g., User, Grievance Officer, Administrator).
- **Key Columns:** RoleID (PK), RoleName, Description.

3. Departments (Organizational Structure)

- **Description:** Stores the various departments within the organization (e.g., IT, HR, Finance).
- **Key Columns:** DepartmentID (PK), DepartmentName, Description.

4. Categories (Grievance Classification)

- **Description:** The predefined categories for grievances (e.g., Network Issue, Payroll, Facility Maintenance).
- **Key Columns:** CategoryID (PK), CategoryName, DepartmentID (FK - indicating which department handles this category).

5. Grievances (Core Transactional Data)

- **Description:** The central table storing all submitted grievances.
- **Key Columns:** GrievanceID (PK), SubmitterUserID (FK), Title, Description, Status (e.g., Submitted, Under Review, Resolved), CategoryID (FK - Final category), Priority (e.g., Low, Medium, High, Critical), AssignedOfficerID (FK), CreatedAt, UpdatedAt, ResolvedAt.

6. GrievanceAIRecommendations (AI Outputs)

- **Description:** Stores the raw recommendations provided by the AI module before human officer confirmation. This helps in tracking AI accuracy over time.
- **Key Columns:** RecommendationID (PK), GrievanceID (FK), PredictedCategoryID (FK), PredictedPriority, ConfidenceScore, RecommendationDate.

7. GrievanceHistory (Audit Trail)

- **Description:** Tracks every state change and significant update to a grievance for audit and compliance.
- **Key Columns:** HistoryID (PK), GrievanceID (FK), ActionTaken (e.g., Status Changed, Reassigned), OldValue, NewValue, ChangedByUserID (FK), ChangeDate, Comments.

Junction Tables (Mapping Tables)

8. SimilarGrievances (Many-to-Many Relationship)

- **Description:** A junction table that maps grievances to other grievances that the AI has identified as similar or duplicates.
- **Key Columns:** SimilarityID (PK), PrimaryGrievanceID (FK to Grievances), SimilarGrievanceID (FK to Grievances), SimilarityScore (Float), IdentifiedDate.

9. GrievanceAttachments (One-to-Many / Mapping)

- **Description:** If users are allowed to upload files (images, documents) with their grievances.
- **Key Columns:** AttachmentID (PK), GrievanceID (FK), FilePath, FileName, UploadedAt.

Entity Relationship Summary:

- A User belongs to one Department and has one Role.
- A Grievance is submitted by one User, assigned to one Officer (User), and belongs to one Category.
- A Grievance has many GrievanceHistory records.
- A Grievance can have one or more GrievanceAIRecommendations.
- The SimilarGrievances table links a Grievance to multiple other Grievances based on AI similarity scores.

